**5** **(a)** The variation with time $t$ of the potential difference $V_1$ across a resistor is shown in Fig. 5.1.

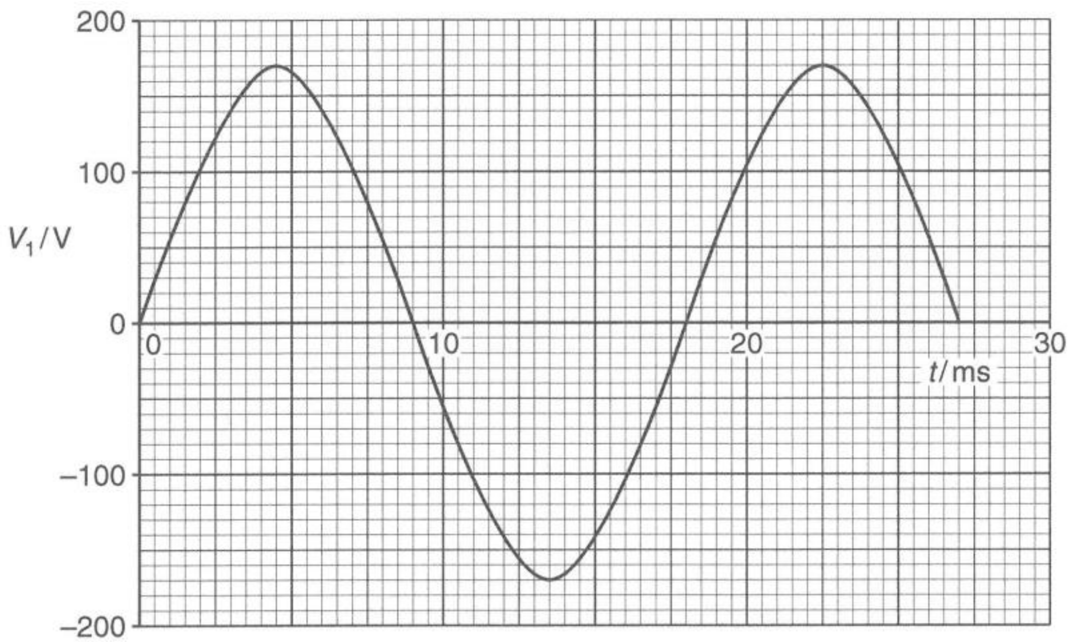


**Fig. 5.1**

The relation between $V_1$ and $t$ is given by

$$V_1 = V_0 \sin \omega t \, .$$

Use Fig. 5.1 to determine the root-mean-square voltage of $V_1$.

Mean-square voltage $= \langle V^2 \rangle = \frac{1}{2} V_0{}^2 = \frac{1}{2} \times 170^2 = 14450$

Root-mean-square voltage $= \sqrt{\langle V^2 \rangle} = \sqrt{14450}$ = 120 V

root-mean-square voltage = .......................................... V [1]

**(b)** The potential difference $V_1$ shown in Fig. 5.1 is connected to an ideal transformer, as shown in Fig. 5.2. The primary coil has 500 turns and the secondary coil has 20 turns. The secondary coil is connected to an open switch and a 15 Ω resistor.

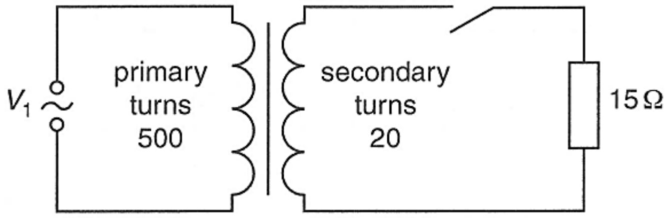


**Fig. 5.2**

The switch in the secondary circuit is now closed.

Determine

**(i)** the peak current in the 15 Ω resistor,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Comparing peak voltages,

$$\frac{V_s}{170} = \frac{20}{500}$$

Peak secondary voltage = $V_s$ = 6.8 V

Peak current = $I = \frac{V}{R} = \frac{6.8}{15}$ = <u>0.45 A</u>

peak current = ........................................ A [2]

**(ii)** the mean power dissipated in the 15 Ω resistor.

r.m.s. secondary voltage = $V_{rms} = \frac{V_0}{\sqrt{(2)}} = \frac{6.8}{\sqrt{(2)}}$ = 4.8 V

Mean power dissipated in resistor = $\langle P \rangle = \left\langle \frac{V^2}{R} \right\rangle = \frac{{V_{rms}}^2}{R} = \frac{4.8^2}{15}$ = <u>1.5 W</u>

mean power dissipated = ........................................ W [2]

**(c)** For a non-ideal transformer, suggest why thermal energy is generated in the soft iron core when the transformer is in use.

The magnetic flux generated by the primary coil changes with time continuously. [1] By Faraday's Law, this means that there is an induced e.m.f. within the soft iron core. This induced e.m.f. results in eddy currents in the core that produce heat due to joule heating. [1]

[2]

[Total:7]

6 A single-turn copper square frame of length $L$ is rotated with constant angular speed $\omega$ by an